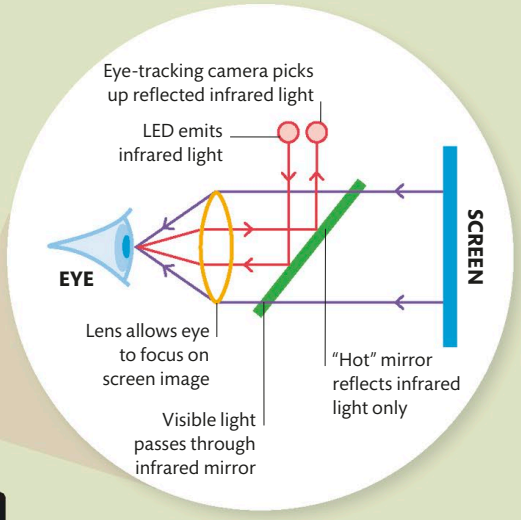
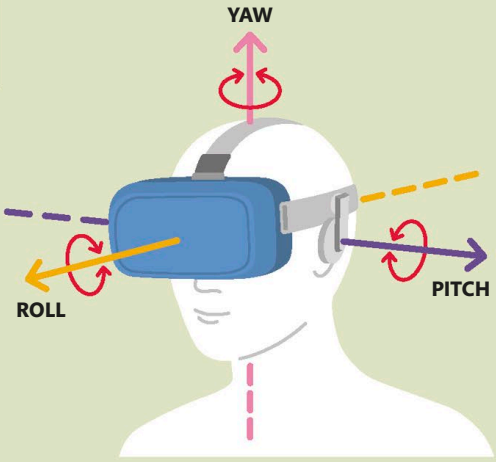
Virtual reality

Our brains perceive the world around us because they receive information from our senses—in particular, our eyes and our ears. By feeding our senses with sights and sounds generated inside a computer, via a virtual reality headset, our brains can perceive worlds that do not really exist—virtual worlds.



Eye tracking
Some VR headsets detect eye movements by shining invisible infrared radiation at the eyes. A mirror inside diverts the infrared to a sensor that can track the eye movements.





Head tracking
Inside a VR headset is a device called an accelerometer (see p.207), which detects head movements. The computer adjusts the view of the virtual world accordingly, so the user can look around the virtual world.

How a VR headset works

A virtual reality (VR) headset displays two views of the virtual world—one for each eye. This gives a sense of depth so that virtual objects appear at different distances, enhancing the feeling of presence. The headset detects the user's head position and movements—and, in some cases, eye movements—and feeds this information to the computer, which adjusts the view and allows the user to look around in the virtual world. Most headsets also contain stereo headphones so sounds from the virtual world can be heard.



OMNIDIRECTIONAL TREADMILLS ARE BEING DEVELOPED SO VR USERS CAN **WALK FREELY IN VIRTUAL WORLDS**